

Chapter

The Circulatory System

1. During ventricular systole, the atrioventricular valves (P) and semilunar valves (Q) SQP 2025

- (a) P — close, Q — Open
- (b) P — close, Q — Close
- (c) P — open, Q — Close
- (d) P — open, Q — Open

2. The mineral element in hemoglobin : SQP 2021

- (a) Manganese
- (b) Iron
- (c) Sodium
- (d) Calcium

3. The blood vessel that carries oxygenated blood is : MAIN 2021

- (a) Pulmonary artery
- (b) Pulmonary vein
- (c) Renal vein
- (d) Hepatic vein

**4. The nearest organ to which the heart supplies oxygenated blood is
COMP 2021**

- (a) Kidney
- (b) Heart

- (c) Lung
- (d) Intestine

5. The opening of right auricle into the right ventricle is guarded by

SQP 2020

- (a) Bicuspid valve
- (b) Tricuspid valve
- (c) Aortic semi-lunar valve
- (d) Pulmonary semi-lunar valve

6. Which of the following cells do not exhibit phagocytic activity? MAIN 2021

- (a) Monocytes
- (b) Neutrophil
- (c) Macrophage
- (d) Basophil

7. Rate of heart beat is controlled by

COMP 2022

- (a) Cranial nerves
- (b) SA node
- (c) Sympathetic nerves
- (d) Spinal nerves

8. Double circulation means

MAIN 2023

- (a) Every organ has an artery and a vein.
- (b) Blood flows twice through the heart.
- (c) Oxygenated and deoxygenated blood is separated.
- (d) Both (a) and (b)

9. Which of the following chambers of heart first receives blood from the systemic circulation? COMP 2023

- (a) Left ventricle
- (b) Right ventricle
- (c) Left auricle
- (d) Right auricle

10. The nearest organ to which the heart supplies oxygenated blood is COMP 2023

- (a) Stomach
- (b) Lung
- (c) Intestine
- (d) Heart itself

11. During a heart examination, a doctor explains that a specific node in the heart generates electrical impulses that regulate the heart's rhythm, acting as the natural pacemaker. The pacemaker of the heart is: MAIN 2021

- (a) AV node
- (b) SV node
- (c) SA node
- (d) tricuspid valve

12. A doctor explains that a type of agranulocyte is crucial for the body's immune response, as it produces antibodies and helps fight infections. This agranulocyte is: SQP 2021

- (a) Basophils
- (b) Neutrophils
- (c) Lymphocytes

(d) Eosinophils

13. The smallest blood vessels in the human body. SQP 2025
14. Irregular-shaped blood cells are____. SQP 2023
15. A straw coloured fluid connective tissue without RBCs. COMP 2022
16. The phase of the cardiac cycle in which the auricle contact. MAIN 2023
17. The phase of cardiac cycle in which the ventricles relax. COMP 2021
18. The vitamin necessary for blood clotting. COMP 2023
19. A genetic disorder in which blood does not clot. SQP 2022
20. An unstable bright red compound formed between Hemoglobin and oxygen to carry the oxygento the body cell. SQP 2022
21. What are the two types of blood circulation in humans ? COMP 2021
22. Lymphocytes COMP 2021
23. Thrombocytes MAIN 2021
24. Hemoglobin MAIN 2022
25. Mammals have open circulatory system. SQP 2023
26. Tricuspid valve COMP 2023
27. The pulmonary artery contains deoxygenated blood. MAIN 2023
28. Write any two functions of Lymph. SQP 2025
29. Prothronibin, Thrombin, Fibrinogen, Albumin MAIN 2024
30. Mention two structural differences between an artery and a vein.
MAIN 2024
31. Why are RBCs efficient in their functions though they lack nucleus and mitochondria? MAIN 2024
32. A mature mammalian erythrocyte lacks nucleus and mitochondria but is efficient in its functioning. Explain by giving suitable reasons. COMP 2021

33. Why are pulmonary circulation and systemic circulation called so?

SQP 2023

34. What is Plasma ? What are the various functions of Plasma ? SQP 2021

Solution

1. (a) P — close, Q — open

2. (b) Iron

Iron is an essential mineral present in hemoglobin, the protein in red blood cells that carries oxygen throughout the body.

3. (b) Pulmonary vein

Pulmonary veins carry **oxygenated blood** from the lungs to the left atrium of the heart, unlike pulmonary arteries, which carry deoxygenated blood to the lungs.

4. (b) Heart, not lung.

The heart **supplies oxygenated blood to itself** through the **coronary arteries**, which are the first branches of the aorta.

5. (b) Tricuspid valve

The opening between the **right atrium (auricle)** and **right ventricle** is guarded by the **tricuspid valve**, which has **three flaps** to prevent backflow of blood into the atrium during ventricular contraction.

6. (d) Basophil

Basophils **do not exhibit phagocytic activity**. Phagocytosis is carried out by cells like **monocytes, neutrophils, and macrophages**, which ingest and destroy pathogens. Basophils mainly release **histamine and other chemicals** during allergic and inflammatory responses.

7. (b) SA node

The **SA (sinoatrial) node**, located in the right atrium, is the **natural pacemaker of the heart**. It generates electrical impulses that control the **rate and rhythm**

of heartbeats. While the autonomic nervous system can **modify** the heart rate, the SA node **initiates** it.

8. (d) Both (a) and (b)

Double circulation means that blood passes **twice through the heart** in one complete cycle:

1. **Pulmonary circulation** – from heart to lungs and back.
2. **Systemic circulation** – from heart to the rest of the body and back.

9. (d) Right auricle (Right atrium)

The **right auricle (atrium)** is the first chamber of the heart to receive **deoxygenated blood** from the systemic circulation via the **superior and inferior vena cava**. This blood then moves to the **right ventricle** to be pumped to the lungs for oxygenation.

10. (d) Heart itself

The heart supplies oxygenated blood to **itself first** through the **coronary arteries**, which branch directly from the aorta. This ensures the heart muscles receive oxygen and nutrients before blood is delivered to other organs.

11. (c) SA node

The **SA (sinoatrial) node**, located in the right atrium, acts as the **natural pacemaker of the heart**. It generates **electrical impulses** that initiate each heartbeat and regulate the heart's rhythm.

12. (c) Lymphocytes

Lymphocytes are **agranulocytes** that play a key role in the **immune response**. They produce **antibodies** and help the body fight infections. Basophils, neutrophils, and eosinophils either have granules or function differently in immunity.

13. The smallest blood vessels in the human body.

14. WBC

15. Lymph

16. Atrial diastole

17. Ventricular diastole.

18. Vitamin K

19. Hemophilia

20. Oxy Hemoglobin

21. The two types of blood circulation in humans are:

(i) **Pulmonary circulation**

(ii) **Systemic circulation**

22. Produce antibodies

23. Clotting of blood

24. Hemoglobin is a **respiratory pigment** present in red blood cells that **carries oxygen (O₂)** from the lungs to various parts of the body.

25. Mammals have a **closed circulatory system**, where blood flows through blood vessels. **Insects** are the ones with an **open circulatory system**, where blood flows freely into body cavities.

26. Tricuspid valve

The **tricuspid valve** is located between the **right atrium (auricle)** and **right ventricle**. It **prevents the backflow of blood** from the right ventricle into the right atrium during ventricular contraction.

27. True

The **pulmonary artery** carries **deoxygenated blood** from the **right ventricle** of the heart to the lungs for oxygenation.

28. Two functions of lymph are:

1. **Absorption of fats** from the intestines.

2. **Drains excess tissue fluid** back into the blood and **protects the body** from disease-causing germs.

29. Odd One Out: Albumin

Reason: The others (Prothrombin, Thrombin, Fibrinogen) are clotting factors, whereas Albumin is a plasma protein not involved in blood clotting.

30. Differences between artery and vein:

Artery	Vein
They have thicker walls.	They have thinner walls.
They lack valves.	They contain valves to prevent the backflow of blood.

31. Loss of nucleus: Makes RBCs biconcave, increasing their surface area-to-volume ratio, which helps them absorb more oxygen.

Lack of mitochondria: RBCs do not use the oxygen they carry, so all the oxygen absorbed from the lungs is delivered efficiently to body tissues.

32. Loss of nucleus: Increases the surface area-to-volume ratio, allowing the erythrocyte to absorb more oxygen.

Loss of mitochondria: Prevents the cell from using the oxygen it carries, so all the oxygen is delivered to the tissues efficiently.

33. Pulmonary circulation: Blood flows from the right side of the heart to the lungs and back to the left side of the heart.

Systemic circulation: Blood flows from the left side of the heart to all body parts and returns to the right side of the heart.

34. Plasma: A colourless fluid in which blood cells are suspended. It is about 90% water and 10% dissolved substances like proteins, fats, minerals, glucose, antibodies, urea, and hormones.

- **Functions of Plasma:**

1. Transports digested food and metabolic wastes.
2. Carries oxygen and carbon dioxide.

3. Transports hormones throughout the body.
4. Helps distribute body heat and maintain body temperature.
5. Maintains water balance in cells.
6. Helps maintain a steady internal environment (homeostasis).